

Superstore Marketing Campaign Analysis

Business Context

Every year-end, the Superstore chain runs a massive promotional campaign and invites its existing customers to buy a gold membership offering a 20 % discount on all purchases.

To make this initiative a success the retailer collects detailed information on every shopper—demographics, purchasing habits, channel preferences and recency. The campaign dataset contains about 2 200 customers and 22 features, including the Response flag that shows whether each customer accepted the membership offer.

This dataset gives us the opportunity to discover what drives membership uptake, identify profitable customer segments and design future marketing strategies.

Project Objective

Management wants to know whether its marketing spend is paying off and how to improve the next campaign. Specifically, the project should:

- **Segment customers** based on demographic attributes (age, education, income, marital status, household composition) and spending patterns.
- **Explore behavioural drivers** such as recency, product category spend (wine, meat, fruits, fish, sweets, gold), and channel usage (web, store, catalogue, discounted purchases). Assess how these factors influence total spend and membership acceptance.
- **Predict membership acceptance** by building classification models (e.g. logistic regression, decision trees, random forest). The Response column is the target variable.
- **Identify high-value and at-risk segments** to support targeted marketing and loyalty programs.
- **Recommend actionable strategies** to increase membership uptake, cross-sell complementary products and optimise marketing spend.

Suggested Project Scope & Tools

Tool	Applications
Excel	Exploratory summaries, scenario modeling, hypothesis tracking
SQL (optional)	Data extraction, joins, aggregations from relational databases
Python	Data wrangling, EDA, anomaly detection, visualization, advanced analysis
Statistics	Probability modeling, hypothesis testing, risk scoring, inferential analytics
Power BI / Tableau	Executive dashboard with KPIs, visual analytics, and filters

Project Deliverables

1. Business Understanding & KPI Framework

- Translate the marketing objective (increasing uptake of the Gold Membership) into analytical questions: Who is most likely to accept? Which demographics and channels drive the highest spend and engagement?
- Define and document KPIs: overall response rate (Response), average spend per customer (MntWines, MntMeatProducts, etc. summed), recency of purchase (Recency), average income, average number of web/store/catalog purchases, and complaint rate (Complain).
- Plan segment cuts by age (derived from Year_Birth), education level (Education), marital status (Marital_Status), household composition (Kidhome, Teenhome), income brackets (Income), and customer tenure (Dt_Customer).

2. Data Acquisition & Validation

- Import the CSV using Excel, SQL or Python (pandas) and verify column types and ranges for the 22 fields.
- Check uniqueness of the ID column, completeness of Income, Year_Birth and the various spending and purchase count variables. Identify missing values or anomalies (e.g., zero or negative spends).
- Examine and handle missing values in income or spending columns, convert date strings in Dt_Customer to tenure, and check consistency between Kidhome/Teenhome totals and likely household sizes.
- Standardise categorical features (e.g., consolidate rare categories in Marital_Status like "Absurd," ordinal-encode Education levels if needed).

3. Exploratory Data Analysis & Descriptive Statistics

- Generate distribution plots and summary statistics for key metrics: customer age, income, recency, total spend by product category (MntWines, MntMeatProducts, MntFishProducts, MntFruits, MntSweetProducts, MntGoldProds), and purchase counts by channel (NumDealsPurchases, NumWebPurchases, NumCatalogPurchases, NumStorePurchases, NumWebVisitsMonth).
- Analyse acceptance rates across demographic groups and channels (e.g. compare Response by education level or number of web purchases).
- Identify high-spending customer segments and underpenetrated groups (e.g. older customers who buy heavily but rarely engage online).
- Summarise category-level contribution to total spend and detect outliers in spending behaviour.

4. Customer & Product Segmentation Analysis

- Segment customers by demographic and behavioural attributes: income tiers, age groups, education, marital status, family size, tenure with the company, recency and channel usage patterns.
- Explore product affinities by analysing which customers spend across multiple product categories or purchase through multiple channels.
- Develop customer profiles: loyal high spenders (high total spend and many store/web purchases), bargain hunters (many deals purchases), digital shoppers (frequent web visits), and dormant or at-risk customers (high recency and low spend).
- Compare spending patterns and response rates across segments to identify which groups are more receptive to the membership offer and which may require tailored incentives.

5. Performance Dashboarding (Power BI / Tableau)

- Design interactive dashboards for executives and marketers to monitor campaign performance.

- Visualise KPIs: response rate over time, spend by product category and channel, income distribution, age distribution, household composition, tenure, and complaint rates.
- Create filters and drill-downs by demographic segment, response outcome and purchasing behaviour so users can explore, for example, how different income bands perform by channel or how education level influences response.

6. Strategic Insights & Recommendations

- Summarise the key findings in an executive report supported by charts and KPIs.
- Highlight the top insights—for example, “Customers with income above \$60k and at least three web purchases have a 30 % higher likelihood of accepting the membership,” or “Households with teenagers are significantly underrepresented among respondents.”
- Provide actionable recommendations: develop targeted email campaigns for web-savvy high-income customers, offer cross-sell bundles to those who already spend on multiple categories, re-engage high-recency/low-spend customers with special catalogue offers, and refine the Gold Membership pricing or benefits for segments with low acceptance.

List of Datasets

Table	Description
Superstore Campaign dataset Sales	This dataset captures customer demographic details, household composition, purchasing behaviour across multiple product categories, channel usage patterns, and response to a Gold Membership marketing campaign. It enables segmentation, response prediction, and insight generation for targeted marketing strategies.

Data Tables Provided

Column Name	Description
Id	Unique customer identifier
Year_Birth	Year of birth (used to derive customer age)
Education	Education level (e.g. Basic, Graduation, Master, PhD)
Marital_Status	Marital status (e.g. Single, Married, Together, Absurd, Widow)
Income	Yearly household income in monetary units
Kidhome	Number of children in the household
Teenhome	Number of teenagers in the household
Dt_Customer	Date the customer first enrolled with the company
Recency	Number of days since the customer’s last purchase
MntWines	Amount spent on wine products in the last two years
MntFruits	Amount spent on fruit products in the last two years

Column Name	Description
MntMeatProducts	Amount spent on meat products in the last two years
MntFishProducts	Amount spent on fish products in the last two years
MntSweetProducts	Amount spent on sweet products in the last two years
MntGoldProds	Amount spent on gold products in the last two years
NumDealsPurchases	Number of purchases made with a discount
NumWebPurchases	Number of purchases made via the website
NumCatalogPurchases	Number of purchases made via catalog
NumStorePurchases	Number of purchases made in physical stores
NumWebVisitsMonth	Number of visits to the company's website in the last month
Complain	1 if the customer complained in the last two years, else 0
Response	Target variable: 1 if the customer accepted the gold offer, 0 otherwise

SUGGESTED INSTRUCTIONS

Learners are encouraged to apply their own ideas while executing the project. The following is intended solely as a guideline.

Note: Learners are supposed to check data integrity and handle any inconsistencies accordingly.

Hint: Certain columns may occasionally be missing (MntWines, MntFruits, MntMeatProducts, etc. or demographic fields like Income). Learners must decide whether to impute, group under "Unknown," or treat separately.

Step 1: Business Understanding & Analytical Planning

- **Context and goals:** Explain that the retailer is preparing a holiday sale and offering a "Gold Membership" at a discounted fee. The goal is to predict which customers will accept the membership and to understand spending behaviour across product categories and channels.
- **Challenges:** List issues such as fragmented customer behaviour, differences in category performance, and the need to measure marketing effectiveness.
- **KPIs:** Define key performance indicators: membership response rate (based on Response), total and average customer spend (sum of MntWines, MntMeatProducts, MntFishProducts, MntFruits, MntSweetProducts and MntGoldProds), category penetration, repeat-purchase proxies (e.g. high NumWebVisitsMonth with low Recency) and the share of high-value segments.
- **Hypotheses:** Formulate testable statements such as "Customers with higher Income are more likely to accept the membership" or "Recent purchasers with high NumWebPurchases have a higher acceptance rate."

Step 2: Data Validation Using Excel

- **Integrity checks:** Verify that each row has a unique ID and that required fields (Response, Income, Year_Birth, Dt_Customer) are present.
- **Highlight issues:** Use conditional formatting to flag missing or zero values in monetary columns (MntWines, MntMeatProducts, etc.) and count columns (NumDealsPurchases, NumWebPurchases, NumCatalogPurchases, NumStorePurchases, NumWebVisitsMonth), and to spot implausible Year_Birth entries.
- **Standardisation:** Ensure consistency across categorical fields (Education, Marital_Status) and record anomalies and suggested corrections in a data quality log.

Step 3: Database Design and Table Creation (SQL DDL)

- **Schema definition:** Design a table (e.g. superstore_campaign) with columns for ID, Year_Birth, Education, Marital_Status, Income, Kidhome, Teenhome,

Dt_Customer, Recency, each Mnt* spending column, each Num* purchase count column, Complain and Response.

- **Constraints:** Specify primary and unique keys (ID), add NOT NULL constraints where appropriate, and include CHECK constraints to ensure monetary and count values are non-negative.
- **Optional views:** Create dimension-like views such as customer_info (demographics and tenure), spending_profile (spend columns plus total), and channel_usage (purchase counts).

Step 4: Data Ingestion into SQL

- **Import data:** Load the cleaned CSV into the superstore_campaign table using a bulk import command.
- **Verify counts:** Confirm that the number of rows matches the number of customers and that there are no duplicate ID values.
- **Sanity queries:** Perform simple checks (distinct counts, sums of spend columns, min/max values) to ensure data imported correctly.

Step 5: Connecting Python to SQL

- **Set up environment:** In a Jupyter notebook, import pandas, sqlalchemy and a relevant database driver.
- **Connect:** Create an engine with a secure connection string and read the table into a DataFrame.
- **Validate:** Check dtypes, shapes and sample data in the DataFrame. Run a few queries from Python to replicate basic SQL checks.

Step 6: Data Cleaning, Transformations & Enhanced EDA in Python

- **Handle missing values:** Impute missing Income values, and decide how to treat zero or missing entries in the spending columns.
- **Feature engineering:**
 - Compute Age from Year_Birth.
 - Calculate customer tenure from Dt_Customer.
 - Compute Total_Spend by summing all Mnt* columns.
 - Determine Category_Breadth by counting how many product categories have non-zero spend.
 - Derive engagement metrics from the Num* columns, such as total purchases or ratios of web vs. store purchases.
- **Exploratory analysis:** Use histograms, boxplots and correlation matrices to explore distributions and relationships. Identify high-spending and under-penetrated segments.

Step 7: Statistical Analysis in Python

- **Group comparisons:** Use t-tests or non-parametric alternatives to compare spending and acceptance rates across different groups (e.g. Education, Marital_Status, households with Kidhome and/or Teenhome).

- **Variance analysis:** Apply ANOVA or Kruskal–Wallis tests to analyse differences across multiple categories (e.g. education levels, income brackets).
- **Relationship tests:** Use chi-square tests to examine associations between categorical variables (e.g. Education vs. Response).
- **Modelling:** Fit logistic regression, decision tree or random forest models to predict Response, and interpret feature importance to understand the drivers of acceptance.

Step 8: Dashboard Development (Power BI / Tableau)

- **Connect:** Link your BI tool to the SQL database.
- **Dashboard types:**
 - **Executive Dashboard:** Include metrics such as response rate, total and average spend, top categories by Mnt* columns, purchase channel usage, and demographic breakdowns. Use filters for Age, Income, Education, Marital_Status, Kidhome and Teenhome.
 - **Marketing Dashboard:** Focus on segment performance and predictors of acceptance. Use scatter plots and heatmaps to show relationships such as Total_Spend vs. Recency, with slicers for channel usage and product category breadth.
- **Interactivity:** Provide drill-down capabilities so stakeholders can slice data by specific segments and view corresponding KPIs.

Step 9: Strategic Insights & Recommendations

- **Summarise key findings:** Highlight patterns discovered in the analysis—for example, high-income customers with many NumWebPurchases show greater acceptance, households with teenagers have lower response rates, or particular Mnt* categories are strong predictors of loyalty and response.
- **Actionable recommendations:** Suggest targeted marketing campaigns for high-income digital shoppers, family-oriented offers for households with children, and retention programmes for high-spending non-responders. Justify each recommendation with data.
- **Report and presentation:** Prepare a concise report and slide deck that outline insights, provide supporting charts and detail the business impact of proposed actions.